G SCALE RAILWAY CONTROL

LADDER DIAGRAM PACK

Raspberry Pi 5 · MCP23017 x3 · BC547 LED Drivers
8-Channel Relay Module · KY-024 Hall Effect Sensors
BTS7960 H-Bridge · ADS1115 Current Sense

DOCUMENT:	Ladder Diagram Pack
REVISION:	Rev A
DATE:	April 2026
SHEETS:	8 (Title + Index + LD-01 to LD-06)
FORMAT:	A4 Portrait
SCALE:	Not to Scale (NTS)

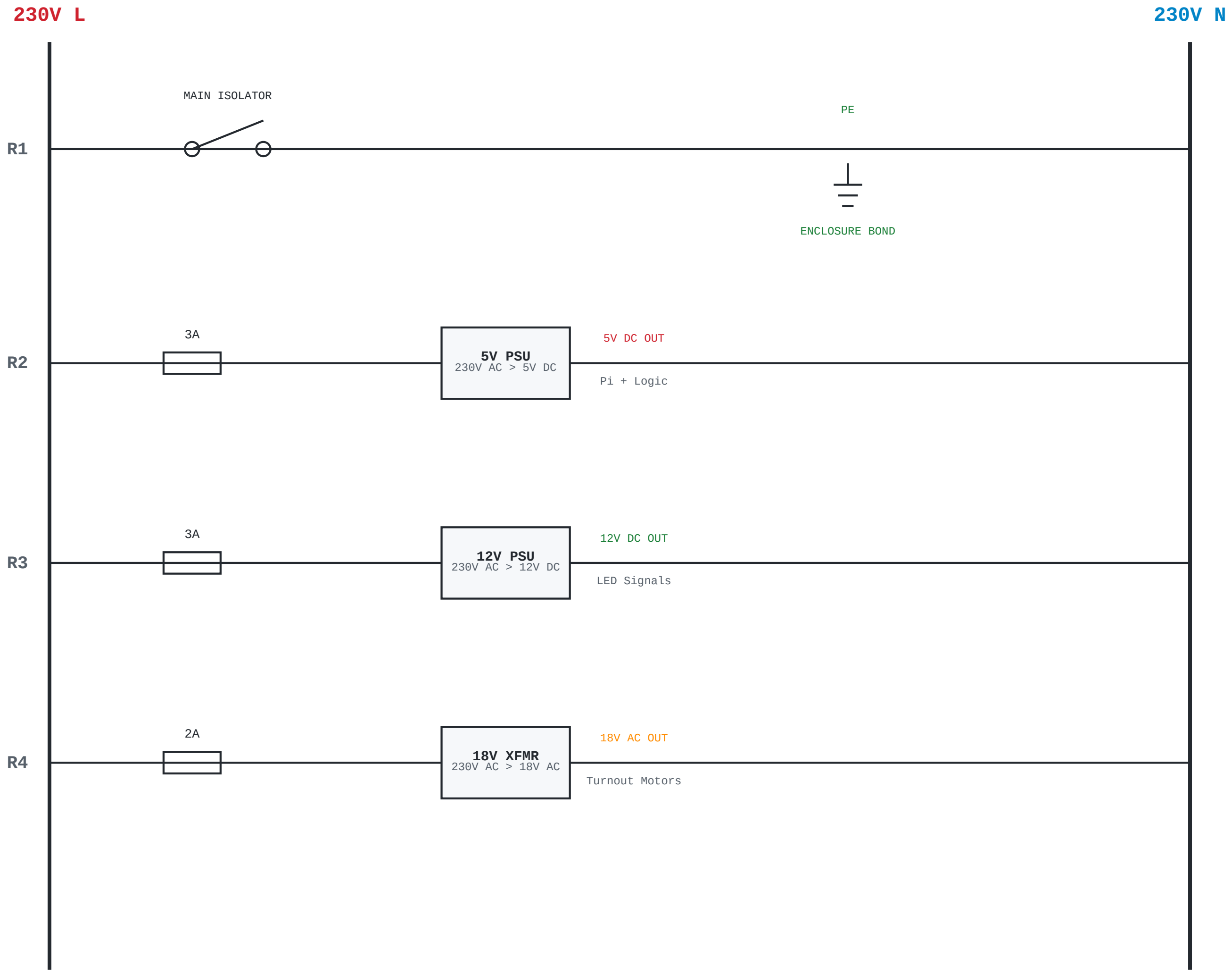
SHEET INDEX

CROSS-REFERENCES

- Terminal block schedules: a3-detailed-subcomponent-sheets.html (Rev C), sheets D01-D10
- System wiring overview: wiring_diagram.html
- Pin assignments (wiring view): rail-control.html — Wiring tab
- Bench testing regime: Stages 0-7 markdown files
- Component BOM: README.md — Bill of Materials

LADDER DIAGRAM CONVENTIONS

- Left vertical rail: Positive supply (+V or L)
- Right vertical rail: Return (0V, N, or GND)
- Horizontal rungs: One circuit per rung, read left to right
- Rung numbering: R1, R2, R3... sequential per sheet
- All component values shown on diagram
- Wire colour codes cross-referenced to A3 drawing pack where applicable



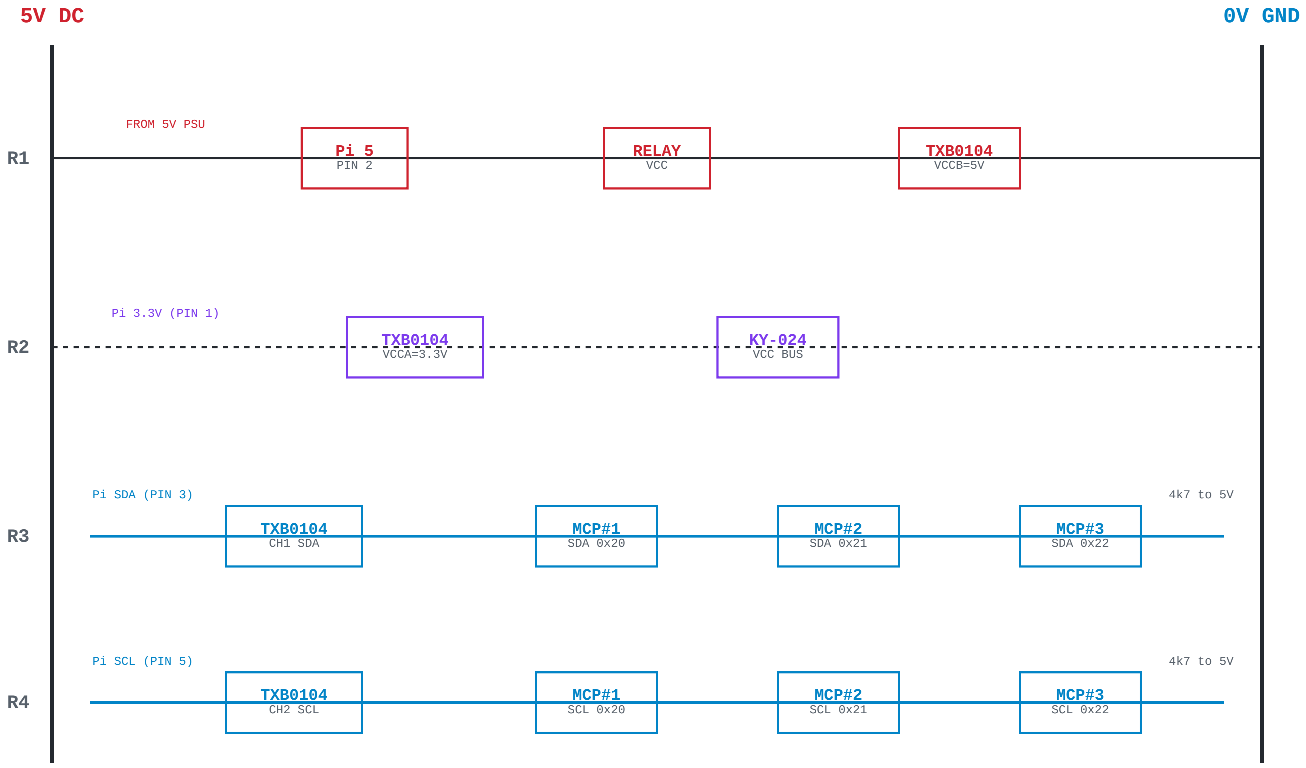
NOTE: 18V AC circuit is galvanically isolated from DC circuits via relay contacts. DC ground bus and AC neutral are NOT connected.
All metalwork bonded to PE via main earth terminal.

DIN RAIL FUSE SCHEDULE

POS	RATING	TYPE	PROTECTED CIRCUIT	NOTES
F1	3A	HRC	5V PSU (Pi, logic, relay module)	Slow-blow
F2	3A	HRC	12V PSU (LED signal drivers)	Slow-blow
F3	2A	HRC	18V AC transformer (turnout motors)	Slow-blow

SUPPLY SUMMARY

SUPPLY	VOLTAGE	TYPE	MIN CURRENT	FEEDS
5V PSU	5V DC	Regulated	2A	Raspberry Pi, relay module logic, TXB0104
12V PSU	12V DC	Regulated	2A	BC547 LED driver circuits (40 outputs)
18V XFMR	18V AC	Transformer	1A	LGB turnout motors via relay contacts



COMMON GROUND BUS

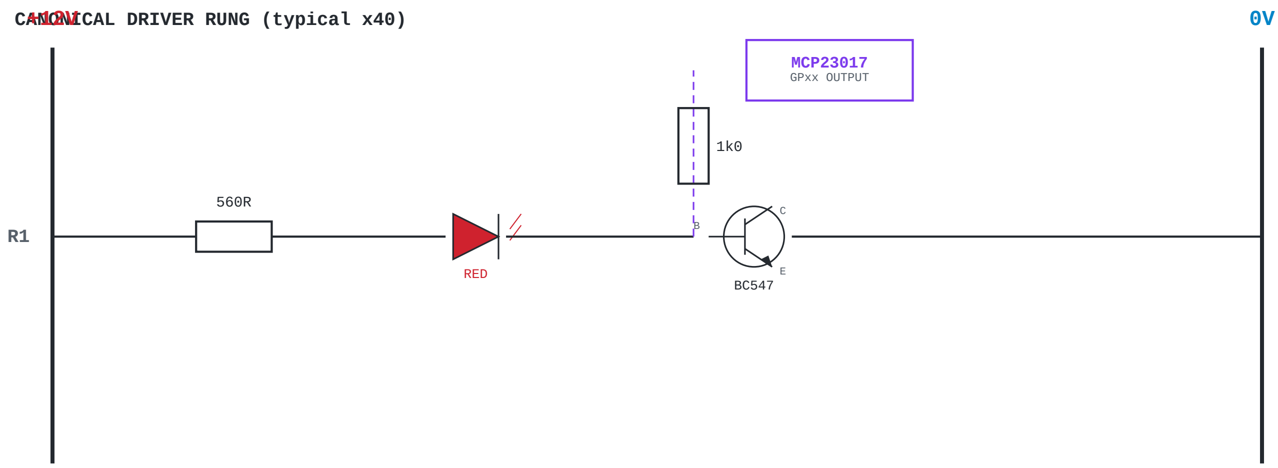
CONNECTION	SOURCE	TERMINAL
Pi GND	PIN 6 (or PIN 9/14/20/25)	TB-GND
5V PSU 0V	PSU negative terminal	TB-GND
12V PSU 0V	PSU negative terminal	TB-GND
MCP23017 #1 GND (VSS)	CJMCU-2317 board #1	TB-GND
MCP23017 #2 GND (VSS)	CJMCU-2317 board #2	TB-GND
MCP23017 #3 GND (VSS)	CJMCU-2317 board #3	TB-GND
Relay module GND	8-ch relay board	TB-GND
TXB0104 GND	Level shifter board	TB-GND
KY-024 sensors GND	All hall effect sensors	TB-GND
BC547 emitter returns	All LED driver circuits	TB-GND

NOTE: 18V AC neutral is NOT connected to this ground bus.
The AC circuit returns via its own neutral to the LGB transformer.

MCP23017 I2C ADDRESS CONFIGURATION

BOARD	A2	A1	A0	ADDRESS	FUNCTION
MCP #1	0	0	0	0x20	LED signals 1-16 (GPA0-GPB7)
MCP #2	0	0	1	0x21	LED signals 17-32 (GPA0-GPB7)
MCP #3	0	1	0	0x22	LEDs 33-40 (GPA0-7) + Relays (GPB0-5)

CANONICAL DRIVER RUNG (typical x40)



CURRENT CALCULATION:

I_LED = (V_supply - V_LED - V_CE(sat)) / R_limit

I_LED = (12V - 2.0V - 0.3V) / 560R = 17.3mA (typical red LED)

V_LED varies by colour: Red ~2.0V, Green ~2.2V, Yellow ~2.1V, White ~3.2V

Base current: I_B = (3.3V - 0.7V) / 1k0 = 2.6mA (BC547 hFE > 100, well into saturation)

PIN ASSIGNMENT TABLE — LED OUTPUTS

MCP	PIN	FUNCTION	COLOUR	LOCATION
#1 (0x20)	GPA0	Signal 1	TBD	Assign per layout
#1 (0x20)	GPA1	Signal 2	TBD	Assign per layout
#1 (0x20)	GPA2	Signal 3	TBD	Assign per layout
#1 (0x20)	GPA3	Signal 4	TBD	Assign per layout
#1 (0x20)	GPA4	Signal 5	TBD	Assign per layout
#1 (0x20)	GPA5	Signal 6	TBD	Assign per layout
#1 (0x20)	GPA6	Signal 7	TBD	Assign per layout
#1 (0x20)	GPA7	Signal 8	TBD	Assign per layout
#1 (0x20)	GPB0	Signal 9	TBD	Assign per layout
#1 (0x20)	GPB1	Signal 10	TBD	Assign per layout
#1 (0x20)	GPB2	Signal 11	TBD	Assign per layout
#1 (0x20)	GPB3	Signal 12	TBD	Assign per layout
#1 (0x20)	GPB4	Signal 13	TBD	Assign per layout
#1 (0x20)	GPB5	Signal 14	TBD	Assign per layout
#1 (0x20)	GPB6	Signal 15	TBD	Assign per layout
#1 (0x20)	GPB7	Signal 16	TBD	Assign per layout
#2 (0x21)	GPA0	Signal 17	TBD	Assign per layout
#2 (0x21)	GPA1	Signal 18	TBD	Assign per layout

Table continues on next page...

MCP	PIN	FUNCTION	COLOUR	LOCATION
#2 (0x21)	GPA2	Signal 19	TBD	Assign per layout
#2 (0x21)	GPA3	Signal 20	TBD	Assign per layout
#2 (0x21)	GPA4	Signal 21	TBD	Assign per layout
#2 (0x21)	GPA5	Signal 22	TBD	Assign per layout
#2 (0x21)	GPA6	Signal 23	TBD	Assign per layout
#2 (0x21)	GPA7	Signal 24	TBD	Assign per layout
#2 (0x21)	GPB0	Signal 25	TBD	Assign per layout
#2 (0x21)	GPB1	Signal 26	TBD	Assign per layout
#2 (0x21)	GPB2	Signal 27	TBD	Assign per layout
#2 (0x21)	GPB3	Signal 28	TBD	Assign per layout
#2 (0x21)	GPB4	Signal 29	TBD	Assign per layout
#2 (0x21)	GPB5	Signal 30	TBD	Assign per layout
#2 (0x21)	GPB6	Signal 31	TBD	Assign per layout
#2 (0x21)	GPB7	Signal 32	TBD	Assign per layout
#3 (0x22)	GPA0	Signal 33	TBD	Assign per layout
#3 (0x22)	GPA1	Signal 34	TBD	Assign per layout
#3 (0x22)	GPA2	Signal 35	TBD	Assign per layout
#3 (0x22)	GPA3	Signal 36	TBD	Assign per layout
#3 (0x22)	GPA4	Signal 37	TBD	Assign per layout
#3 (0x22)	GPA5	Signal 38	TBD	Assign per layout
#3 (0x22)	GPA6	Signal 39	TBD	Assign per layout
#3 (0x22)	GPA7	Signal 40	TBD	Assign per layout

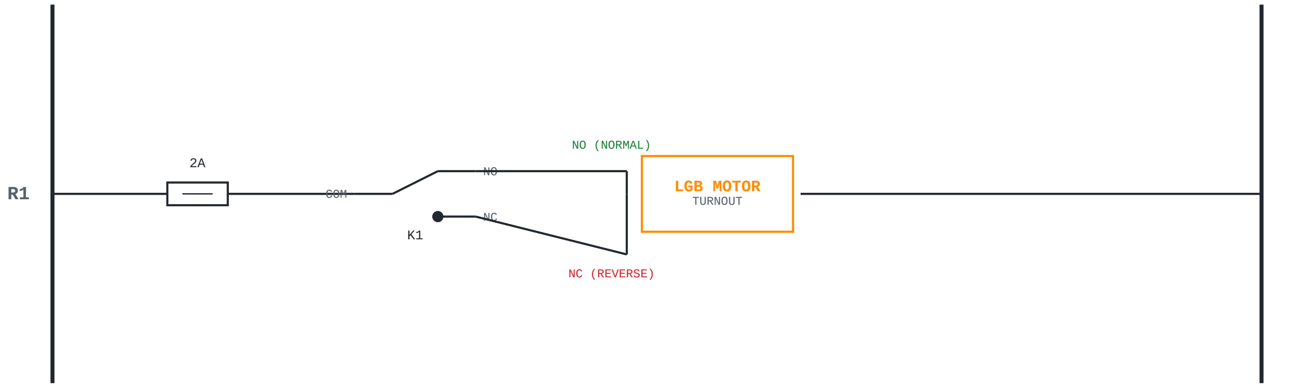
NOTE: All 40 outputs use identical driver circuit (560R + LED + BC547 + 1k0 base).
LED colour and location to be assigned once layout is finalised.
Use high-brightness LEDs for daylight visibility outdoors.

LD-04 18V AC TURNOUT RELAY CIRCUIT

18V AC L

POWER CIRCUIT — RELAY CONTACTS (per turnout)

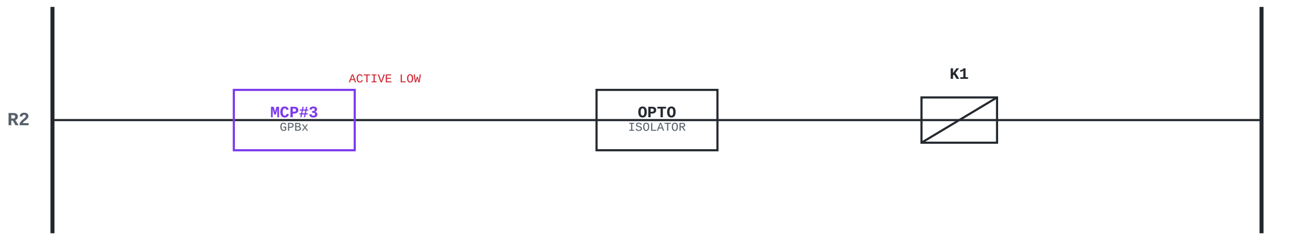
18V AC N



5V (Relay)

COIL DRIVE CIRCUIT (per turnout)

0V GND



WARNING: Never energise both directions simultaneously for the same turnout.
This short-circuits through the motor coil. Add a software interlock.
Pulse timing: 300ms default — adjust per motor. Do NOT hold relay energised.

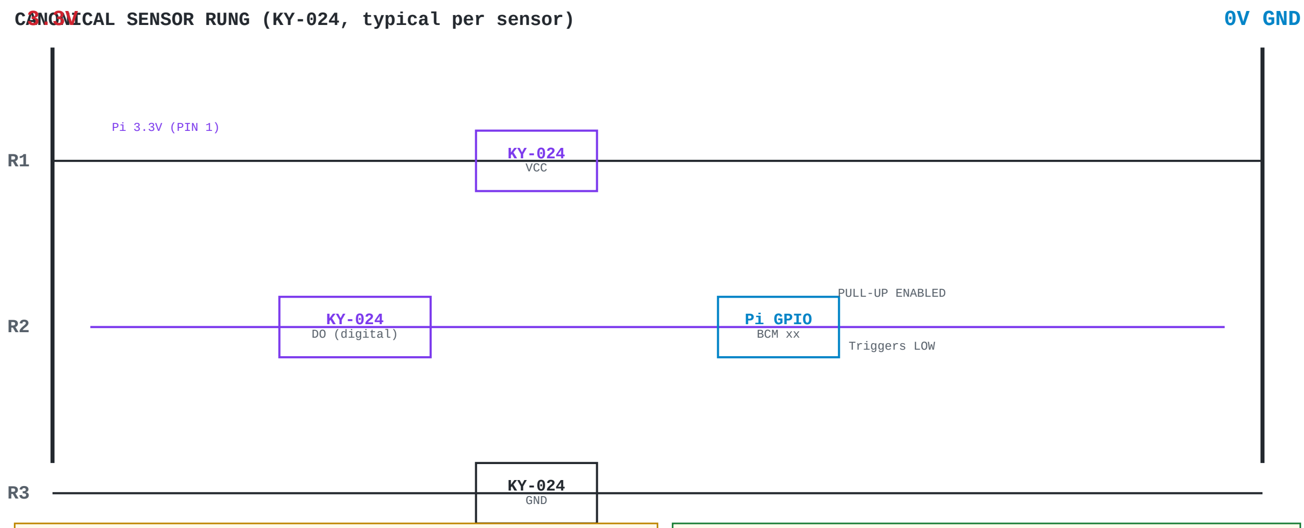
TURNOUT ASSIGNMENT TABLE

RELAY	MCP PIN	TURNOUT ID	LOCATION	NOTES
CH1	#3 GPB0	T1	Assign per layout	300ms pulse
CH2	#3 GPB1	T2	Assign per layout	300ms pulse
CH3	#3 GPB2	T3	Assign per layout	300ms pulse
CH4	#3 GPB3	T4	Assign per layout	300ms pulse
CH5	#3 GPB4	T5	Assign per layout	300ms pulse
CH6	#3 GPB5	T6	Assign per layout	300ms pulse

SOFTWARE PULSE PATTERN:

```
mcp3.get_pin(GPBx).value = False   # Relay ON (active LOW)
time.sleep(0.3)                     # 300ms pulse
mcp3.get_pin(GPBx).value = True    # Relay OFF
```

CANONICAL SENSOR RUNG (KY-024, typical per sensor)



KY-024 AO (analogue output) is NOT USED.
Only the digital output (DO) is connected.
Adjust blue trimmer pot for trigger threshold
with actual loco magnet in position.

OUTDOOR SEALING: Coat PCB connections with
clear silicone or nail varnish. Do NOT coat
the sensor tip (small black component) or it
will not trigger reliably.

SENSOR GPIO ASSIGNMENT (placeholder — finalise with layout)

SENSOR	BLOCK	POSITION	BCM GPIO	CABLE GLAND
S1	BLK-A	Entry	GPIO 4	PG7-01
S2	BLK-A	Exit	GPIO 17	PG7-02
S3	BLK-B	Entry	GPIO 27	PG7-03
S4	BLK-B	Exit	GPIO 22	PG7-04
S5	BLK-C	Entry	GPIO 5	PG7-05
S6	BLK-C	Exit	GPIO 6	PG7-06
S7	BLK-D	Entry	GPIO 13	PG7-07
S8	BLK-D	Exit	GPIO 19	PG7-08
S9	TBD	TBD	GPIO 26	PG7-09
S10	TBD	TBD	GPIO 16	PG7-10

GPIO pins are placeholder assignments — finalise once layout is built in rail-editor.html

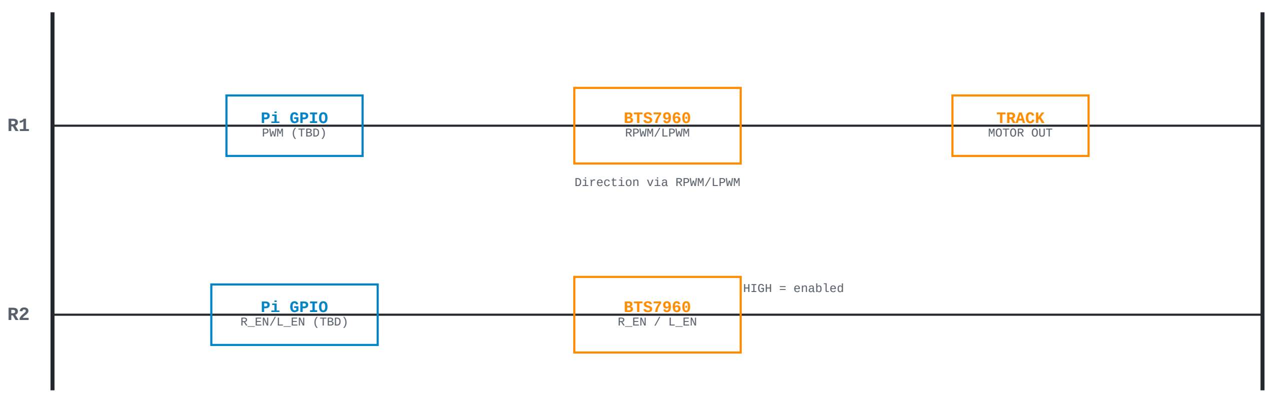
Two sensors per block (entry + exit) enables full block occupancy detection

SOFTWARE EDGE DETECTION:

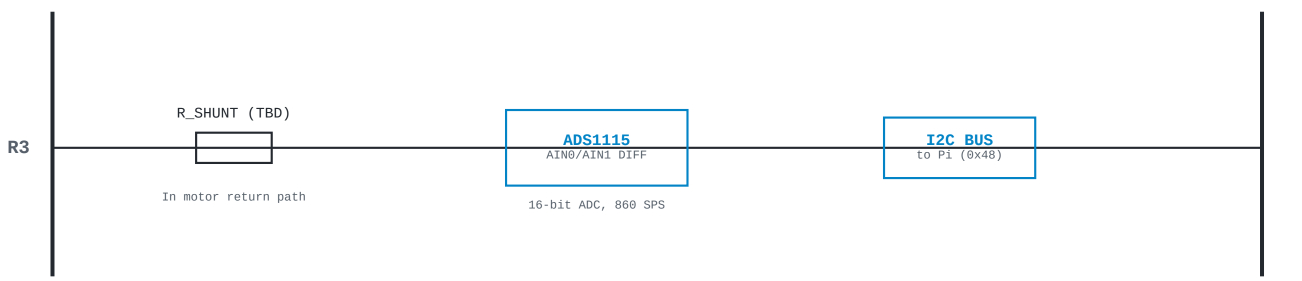
```
GPIO.setup(pin, GPIO.IN, pull_up_down=GPIO.PUD_UP)
GPIO.add_event_detect(pin, GPIO.FALLING, callback=on_train, bouncetime=200)
# Sensor triggers LOW when magnet present (active LOW)
```

THIS SHEET IS A PLACEHOLDER — values marked TBD.
Deferred until outdoor track is laid and traction testing can begin.
See bench testing regime Stage 7+ for commissioning plan.

BTS7960 H-BRIDGE — MOTOR DRIVE



ADS1115 CURRENT SENSE — FEEDBACK LOOP



TRACTION PARAMETERS (ALL TBD)

PARAMETER	VALUE	NOTES
Track voltage	TBD	G Scale nominal — measure from LGB controller
Max traction current	TBD	Depends on loco motor draw
Shunt resistor value	TBD	Size for ADS1115 input range
PWM frequency	TBD	Typical 1-25 kHz for DC motors
ADS1115 I2C address	0x48	Default — ADDR pin to GND
ADS1115 gain setting	TBD	PGA setting depends on shunt voltage range
BTS7960 supply voltage	TBD	Matches track voltage
Pi GPIO for RPWM	TBD	Hardware PWM preferred (GPIO 12/13/18/19)
Pi GPIO for LPWM	TBD	Hardware PWM preferred
Pi GPIO for R_EN	TBD	Standard GPIO sufficient
Pi GPIO for L_EN	TBD	Standard GPIO sufficient